

2021-2022-2 高等数学 I (1)补考卷参考答案

1. 函数 $f(x) = \frac{\ln|1+x|}{(e^x-1)(x-2)}$ 有几个间断点, 并判断是第几类间断点.

解: $f(x)$ 在 $x = -1, x = 0$ 和 $x = 2$ 处无定义, 故有三个间断点. 2'

$$\lim_{x \rightarrow -1} \frac{\ln|1+x|}{(e^x-1)(x-2)} = \lim_{x \rightarrow -1} \frac{\ln|1+x|}{-3(e^{-1}-1)} = \infty,$$

$$\lim_{x \rightarrow 0} \frac{\ln|1+x|}{(e^x-1)(x-2)} = \lim_{x \rightarrow 0} \frac{x}{-2x} = -\frac{1}{2},$$

$$\lim_{x \rightarrow 2} \frac{\ln|1+x|}{(e^x-1)(x-2)} = \lim_{x \rightarrow 2} \frac{\ln 3}{(e^2-1)(x-2)} = \infty, \quad 2'$$

故 $x = 0$ 是第一类间断点中的可去间断点,

$x = -1$ 和 $x = 2$ 是第二类间断点. 2'

2. 已知极限 $\lim_{x \rightarrow 0} \left(a \arctan \frac{1}{x} + (1+2x)^{\frac{1}{|x|}} \right)$ 存在, 求 a 的值

$$\text{解: } \lim_{x \rightarrow 0^+} \left(a \arctan \frac{1}{x} + (1+2x)^{\frac{1}{|x|}} \right) = \lim_{x \rightarrow 0^+} \left(a \arctan \frac{1}{x} + (1+2x)^{\frac{1}{x}} \right) = \frac{\pi}{2}a + e^2 \quad 2'$$

$$\lim_{x \rightarrow 0^-} \left(a \arctan \frac{1}{x} + (1+2x)^{\frac{1}{|x|}} \right) = \lim_{x \rightarrow 0^-} \left(a \arctan \frac{1}{x} + (1+2x)^{\frac{-1}{x}} \right) = -\frac{\pi}{2}a + \frac{1}{e^2} \quad 2'$$

$$\text{由 } \frac{\pi}{2}a + e^2 = -\frac{\pi}{2}a + \frac{1}{e^2} \text{ 得, } a = \frac{1}{\pi} \left(\frac{1}{e^2} - e^2 \right) \quad 2'$$

3. 求极限 $\lim_{x \rightarrow 0} \left(\frac{2+e^x}{3} \right)^{\cot x}$

$$\begin{aligned} \text{解: } & \lim_{x \rightarrow 0} \left(\frac{2+e^x}{3} \right)^{\cot x} \\ &= \lim_{x \rightarrow 0} \left(1 + \frac{e^x-1}{3} \right)^{\cot x} \quad 2' \end{aligned}$$

$$= e^{\lim_{x \rightarrow 0} \frac{e^x-1}{3 \tan x}} \quad 2'$$

$$= e^{\frac{1}{3}} \quad 2'$$

4. 已知函数 $f(x)$ 在 $x = 1$ 处连续, 且 $\lim_{x \rightarrow 1} \frac{f(x)}{\ln x} = 1$, 求 $f'(1)$

$$\text{解: } f(1) = \lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \left(\frac{f(x)}{\ln x} \cdot \ln x \right) = \lim_{x \rightarrow 1} \frac{f(x)}{\ln x} \lim_{x \rightarrow 1} \ln x = 0 \quad 2'$$

$$f'(1) = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{\ln(1+x-1)} = \lim_{x \rightarrow 1} \frac{f(x)}{\ln x} = 1 \quad 4'$$

5. 已知函数 $f(x) = e^{\sin x} - e^{-\sin x}$, 求 $f''(2\pi)$

解法一:

$$f'(x) = e^{\sin x} \cos x + e^{-\sin x} \cos x \quad 2'$$

$$f''(x) = (e^{\sin x} \cos^2 x - e^{\sin x} \sin x) + (-e^{-\sin x} \cos^2 x - e^{-\sin x} \sin x) \quad 2'$$

$$f''(2\pi) = 0 \quad 2'$$

解法二:

$f(x)$ 是周期为 2π 的奇函数, 故 $f''(x)$ 也是周期为 2π 的奇函数, $3'$

$$\text{从而 } f''(2\pi) = f''(0) = 0 \quad 3'$$

6. 设平面区域 D 由曲线 $y = \sqrt{x} \sin \pi x$ ($0 \leq x \leq 1$) 与 x 轴所围成, 求 D 绕 x 轴旋转所成旋转体的体积.

$$\text{解: } V = \pi \int_0^1 (\sqrt{x} \sin \pi x)^2 dx \quad 2'$$

$$= \frac{1}{\pi} \int_0^1 \pi x \sin^2 \pi x d(\pi x) = \frac{1}{\pi} \int_0^\pi t \sin^2 t dt \quad 2'$$

$$= \frac{1}{\pi} \int_0^\pi \sin^2 t dt = \int_0^{\frac{\pi}{2}} \sin^2 t dt = \frac{\pi}{4} \quad 2'$$

7. 已知函数 $y = y(x)$ 由方程 $x^2 + xy + y^3 = 1$ 确定, 求 $y''(0)$.

解: 将 $x = 0$ 代入方程, 得 $y = 1$. $2'$

在方程两边同时求一阶、二阶导数, 得

$$2x + y + xy' + 3y^2 y' = 0 \quad 2'$$

$$2 + y' + (y' + xy'') + [6y(y')^2 + 3y^2 y''] = 0 \quad 2'$$

将 $x = 0$, $y = 1$ 代入得,

$$1 + 3y' = 0, \quad 2 + 2y' + 6(y')^2 + 3y'' = 0 \quad 2'$$

$$\text{解得 } y''(0) = -\frac{2}{3} \quad 2'$$

8. (1) 证明当 $x \in (0,1)$ 时, $\frac{x}{2} < \ln(1+x) < x$;

(2) 利用 (1) 的结论, 比较 $I_1 = \int_0^1 \frac{x}{2(1+\cos x)} dx$, $I_2 = \int_0^1 \frac{\ln(1+x)}{1+\cos x} dx$, $I_3 =$

$\int_0^1 \frac{2x}{1+\sin x} dx$ 的大小.

解: (1) 证明 令 $f(x) = \ln(1+x) - x$, $g(x) = \ln(1+x) - \frac{x}{2}$, 2'

则 $f(0) = g(0) = 0$

当 $x \in (0,1)$ 时, $f'(x) = \frac{1}{1+x} - 1 < 0$, $g'(x) = \frac{1}{1+x} - \frac{1}{2} > 0$, 2'

从而 $f(x) < f(0) = 0$, $g(x) > g(0) = 0$,

即, 当 $x \in (0,1)$ 时, $\frac{x}{2} < \ln(1+x) < x$, 2'

(2) 由 (1) 的结论知, 当 $x \in (0,1)$ 时,

$\frac{x}{2(1+\cos x)} < \frac{\ln(1+x)}{1+\cos x} < \ln(1+x) < x < \frac{2x}{1+\sin x}$, 2'

故 $I_1 = \int_0^1 \frac{x}{2(1+\cos x)} dx < I_2 = \int_0^1 \frac{\ln(1+x)}{1+\cos x} dx < I_3 = \int_0^1 \frac{2x}{1+\sin x} dx$ 2'

9. 设 $f(x)$ 连续, 且 $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 2$, $g(x) = \int_0^{\frac{x}{2}} f(xt) dt$.

(1) 求 $g'(x)$;

(2) 证明 $g'(x)$ 在 $x = 0$ 连续.

解: (1) 由 $f(x)$ 连续, 且 $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 2$, 得

$f(0) = \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{f(x)}{x} \cdot x = 0$.

$g(0) = \int_0^{\frac{1}{2}} f(0) dt = 0$ 2'

$x \neq 0$ 时, $g(x) = \int_0^{\frac{x}{2}} f(xt) dt = \frac{1}{x} \int_0^{\frac{x}{2}} f(xt) dx = \frac{1}{x} \int_0^{\frac{x}{2}} f(u) du$ 2'

$g'(x) = \frac{f(\frac{x}{2})}{2x} - \frac{1}{x^2} \int_0^{\frac{x}{2}} f(u) du$ 2'

$g'(0) = \lim_{x \rightarrow 0} \frac{g(x) - g(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{\int_0^{\frac{x}{2}} f(u) du}{x^2} = \lim_{x \rightarrow 0} \frac{f(\frac{x}{2})}{4x} = \frac{1}{8} \lim_{x \rightarrow 0} \frac{f(\frac{x}{2})}{\frac{x}{2}} = \frac{1}{4}$ 2'

$$\text{所以 } g'(x) = \begin{cases} \frac{f(\frac{x}{2})}{2x} - \frac{1}{x^2} \int_0^{\frac{x}{2}} f(u) du & x \neq 0 \\ \frac{1}{4} & x = 0 \end{cases}$$

$$\begin{aligned} (2) \text{ 由 } \lim_{x \rightarrow 0} g'(x) &= \lim_{x \rightarrow 0} \left(\frac{f(\frac{x}{2})}{2x} - \frac{1}{x^2} \int_0^{\frac{x}{2}} f(u) du \right) \\ &= \lim_{x \rightarrow 0} \frac{1}{4} \frac{f(\frac{x}{2})}{\frac{x}{2}} - \lim_{x \rightarrow 0} \frac{\int_0^{\frac{x}{2}} f(u) du}{x^2} & 2' \\ &= \frac{1}{2} - \frac{1}{4} = \frac{1}{4} = g'(0) \end{aligned}$$

知 $g'(x)$ 在 $x = 0$ 连续 2'

10. 设 $y = y(x)$ 满足 $y'' + 2y' + y = 0$, 且 $y(0) = 0$, $y'(0) = 1$, 求 $\int_0^{+\infty} y(x) dx$.

解: 由特征方程 $r^2 + 2r + 1 = 0$, 得 $r_1 = r_2 = -1$, 2'

所以 $y = (C_1 + C_2 x)e^{-x}$, 2'

代入 $y(0) = 0$, $y'(0) = 1$, 得 $y = xe^{-x}$, 2'

$$\int_0^{+\infty} y(x) dx = \int_0^{+\infty} xe^{-x} dx = -(x+1)e^{-x} \Big|_0^{+\infty} = 1 \quad 4'$$

11. 求微分方程 $y' + \frac{y}{2\sqrt{x}} = 2 + \sqrt{x}$ 满足初始条件 $y(1) = 3$ 的解.

$$\text{解: } y = e^{-\int \frac{1}{2\sqrt{x}} dx} \left[\int (2 + \sqrt{x}) e^{\int \frac{1}{2\sqrt{x}} dx} dx + C \right] \quad 2'$$

$$= e^{-\sqrt{x}} [\int (2 + \sqrt{x}) e^{\sqrt{x}} dx + C] \quad (\text{令 } t = \sqrt{x}) \quad 2'$$

$$= e^{-t} [\int (2 + t) e^t dt^2 + C]$$

$$= e^{-t} [\int (4t + 2t^2) e^t dt + C]$$

$$= e^{-t} [(2t^2 e^t + C)]$$

$$= 2t^2 + Ce^{-t} = 2x + Ce^{-\sqrt{x}} \quad 4'$$

由 $y(1) = 3$ 得 $C = e$,

$$\text{故 } y = 2x + e^{1-\sqrt{x}} \quad (x > 0) \quad 2'$$

12. 设函数 $f(x) \geq 0$ 在 $[1, +\infty)$ 上连续, 若曲线 $y = f(x)$, 直线 $x = 1$, $x = t$ ($t > 1$), 与 x 轴围成的平面图形的面积为 $S(t) = t^2 f(t) - 1$. 求 $y = f(x)$ 所满足的微分方程, 并求该微分方程满足条件 $y|_{x=1} = 1$ 的解.

解: 依题设有 $\int_1^t f(x) dx = t^2 f(t) - 1$, 2'

两边对 t 求导, 得

$$f(t) = 2tf(t) + t^2 f'(t), \quad 2'$$

将上式改写为

$$x^2 y' = (1 - 2x)y, \text{ 此即为 } y = f(x) \text{ 所满足的微分方程,} \quad 2'$$

$$\text{分离变量得 } \frac{dy}{y} = \frac{1-2x}{x^2} dx \quad 2'$$

两边分别积分得,

$$\ln y = -\frac{1}{x} - \ln x^2 + C \quad 2'$$

由题设条件 $y|_{x=1} = 1$ 可得 $C = 1$, 则

$$\ln y = -\frac{1}{x} - \ln x^2 + 1$$

$$\text{故 } y = \frac{1}{x^2} e^{\frac{x-1}{x}} \quad 2'$$